



Basic Programming 2

CPU TIME LIMIT

1 second

MEMORY LIMIT

1024 MB

DIFFICULTY

medium (2.99999999811)

You think you can code?

This problem will test you on various basic programming techniques.

You are given two integers N and t ; and then an array A of N integers (0-based indexing).

Based on the value of t , you will perform an action on A .



t	Action Needed
1	Print “Yes” if there are two integers $x \in A$ and $y \in A$ such that $x \neq y$ and $x + y = 7777$, or “No” otherwise (without the quotes)
2	Print “Unique” if all integers in A are different; or print “Contains duplicate” otherwise (without the quotes)
3	Find and print the integer that appears $> \frac{N}{2}$ times in A ; or print -1 if such integer cannot be found
4	Find and print the median integer of A if N is odd; or print both median integers of A if N is even (separate them with a single space)
5	Print integers in A that fall between a range $[100 \dots 999]$ in sorted order; (print a single space between two integers)

Input



The first line of the input contains an integer N and t ($3 \leq N \leq 200\,000$; $1 \leq t \leq 5$).
The second line of the input contains N non-negative 32-bit signed integers.

Output

For each test case, output the required answer based on the value of t .

Scoring

There are 20 hidden test cases that test various requirements of this problem.
All 20 test cases will be tested.
Each hidden test case worth 5 points (the 5 sample test cases below worth 0 point).

Sample Input 1

```
7 1
1 7770 3 4 5 6 7
```

Sample Output 1

```
Yes
```

Sample Input 2

```
7 2
1 2 3 4 5 6 7
```

Sample Output 2

```
Unique
```

Sample Input 3

```
7 3
1 1 1 1 2 2 2
```

Sample Output 3

```
1
```



Sample Input 4

```
8 4
8 1 4 3 6 7 5 2
```

Sample Output 4

```
4 5
```

Sample Input 5

```
7 5
210 999 1000 543 321 99 777
```

Sample Output 5

```
210 321 543 777 999
```